Week 12 Assignment

Accounting for non-independence: mixed-effects models I

Render this file to PDF and submit both the `.qmd` and the rendered `.pdf` via Canvas before Week 13's lecture (Monday). See the syllabus for the late policy.

```
library(tidyverse)
library(lme4)
library(lmerTest)
library(marginaleffects)
```

Part A: Pairing as a mixed model. Using the built-in `sleep` data (rename `group` to `drug` first):

1. Run a paired t-test.
2. Fit `lmer(extra ~ drug + (1 | ID))`, and use `avg_comparisons()` to get the drug effect. Do the estimates match?

Part B: Repeated measures. Using `ChickWeight` restricted to diets 3 and 4:

3. Plot individual chick trajectories coloured by diet.
4. Fit a random-intercept model `weight ~ Time * Diet + (1 | Chick)`.
5. Use `avg_slopes()` with `by = "Diet"` to estimate growth rates. Which diet leads to faster growth, and by how much (with a 95% CI)?

Part C: Concept.

6. A student measures 5 fish from each of 4 tanks (2 tanks per temperature treatment). What is the experimental unit for the temperature effect? How would you model it?
7. In 2–3 sentences: what does `re.form = NA` change in `predictions()`?